

30점중 25점 획득

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① A rigid spacecraft in circular orbit at altitude of 300 km has principal moments of inertia $I_x=1400, I_y=1700, I_z=1000$ kg-m². It is desired to perform rest to rest maneuver from initial attitude Euler angle $\psi=0.5, \theta=0.3, \phi=0$ (rad) to terminal attitude Euler angle $\psi=0, \theta=0, \phi=0$ (rad). The steering maneuver time is 120 s.

- ① Include gravity gradient moments and make use of nonlinear rotational eqs of motion for spacecraft in circular orbit. Assume the steering maneuver is performed by rotating the spacecraft about constant axis.
- ② Determine the constants axis of rotation for steering maneuver. What are the initial and final values of rotation angle about axis?
- ③ Determine the control torques that achieve the steering maneuver. Plot time responses for 3 control axes, 3 axes.

$I_x=1400, I_y=1700, I_z=1000, \psi_0=0.5, \theta_0=0.3, \phi_0=0, \psi_f=0, \theta_f=0, \phi_f=0, t_f=120$ s, $R=6700=6400+300$ m, $\frac{\mu}{R^3}=0.0015$

④ on orbit rotation EOM (gravity gradient 포함). ⑤ LVH에서의 steering maneuver.

1-a) $R_{EL} = R_z(\phi) R_y(\theta) R_x(\psi)$, $R_0 = R_{EL0} = R_z(-\psi_0) R_y(-\theta_0) R_x(-\phi_0) = \begin{pmatrix} 0.8384 & -0.4744 & 0.2593 \\ 0.4580 & 0.8706 & 0.1417 \\ -0.2935 & 0 & 0.9553 \end{pmatrix}$

이 회전행렬을 오일러 주축/각으로 바꿔나간다.

$$\Phi_0 = \cos^{-1} \left(\frac{\text{tr}(R_0) - 1}{2} \right) = 0.5815$$

$$a = a_0 = \frac{1}{2 \sin \Phi_0} \begin{bmatrix} R_{0,22} - R_{0,33} \\ R_{0,33} - R_{0,31} \\ R_{0,21} - R_{0,12} \end{bmatrix} = \begin{bmatrix} -0.1290 \\ 0.5051 \\ 0.8534 \end{bmatrix} \rightarrow \text{이 축을 중심으로 } \Phi_0 = 0.5815 \text{ 에서 } \Phi_f = 0 \text{ 가리 돌려나간다.}$$

1-b) 문제에서 rest to rest steering maneuver를 LVH에서 고려하라고 했다. about constant axis.

$$\dot{w}(t) = a \dot{\Phi}(t) \rightarrow \begin{cases} \Phi_0 = 0.5815, \Phi_f = 0 \\ \dot{\Phi}_0 = 0, \dot{\Phi}_f = 0 \end{cases} \quad (0 \text{ 는 위축과 같고, f는 주축임})$$

$$\Phi(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3, \quad \Phi(t) = \Phi_0 \rightarrow \Phi_f \left(\frac{t}{t_f} \right)^2 + 2 \Phi_0 \left(\frac{t}{t_f} \right)^3$$

$$\dot{\Phi}(t) = b_1 + 2b_2 t + 3b_3 t^2, \quad \dot{\Phi}(t) = 2b_2 + 6b_3 t$$

$$\vec{w}' = \vec{w} - R_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix}$$

$$\vec{w} = \vec{w}' + R_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix} = a \dot{\Phi}(t) + R_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix} = a b_1 + 2a b_2 t + 3a b_3 t^2 + R_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix}$$

$$\dot{\vec{w}} = \dot{\vec{w}}' + \dot{R}_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix} = a \ddot{\Phi}(t) + \dot{R}_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix} = 2a b_2 + 6a b_3 t + \underbrace{S(w_r)}_{S(a \dot{\Phi})} \cdot R_{EL} \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix}$$

∴ Actuator torque $M_A = I \dot{\vec{w}} + \vec{w} \times I \vec{w}$ (중력 그래디언트 토크), $w(t_f) = \begin{pmatrix} 0 \\ w_0 \\ 0 \end{pmatrix}, w'(t_f) = w''(t_f) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \psi \\ \theta \\ \phi \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.3 \\ 0 \end{pmatrix}, \begin{pmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

그림은 5104 1-b1, 1-b2

② Include gravity gradient moments and make use of linear rotational EOM for spacecraft's circular orbit.

① Determine two impulse solution for pitch control torque that achieves pitch slewing maneuver.

What is the change in pitch rate at each impulse? plot pitch vs time.

② a와 동일, 다만 피치만 풀.요.

③ Determine piecewise constant pitch control torque that achieves pitch slewing maneuver. plot pitch vs time.

④ Determine piecewise constant roll control torque that achieves roll slewing maneuver. plot roll vs time.

⑤ Why is good engineering idea to use linear EOM to determine control torque in a-b-d?

필요한 식 먼저 정리해두자.

$$\text{선형화된 피치의 운동식} \Rightarrow \begin{pmatrix} \Delta \dot{\theta} \\ \Delta \dot{\omega}_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -9.35 \times 10^{-7} & 0 \end{pmatrix} \begin{pmatrix} \Delta \theta \\ \Delta \omega_y \end{pmatrix} + \begin{pmatrix} 0 \\ 5.88 \times 10^{-4} \end{pmatrix} M_y$$

$$\text{선형화된 롤의 운동식} \Rightarrow \begin{pmatrix} \Delta \dot{\phi} \\ \Delta \dot{\omega}_x \\ \Delta \dot{\phi} \\ \Delta \dot{\omega}_z \end{pmatrix} = \begin{pmatrix} 0 & 1 & -1.15 \times 10^{-3} & 0 \\ -1.98 \times 10^{-6} & 0 & 0 & 5.75 \times 10^{-4} \\ 1.15 \times 10^{-3} & 0 & 0 & 1 \\ 0 & -3.45 \times 10^{-4} & 0 & 0 \end{pmatrix} \begin{pmatrix} \Delta \phi \\ \Delta \omega_x \\ \Delta \phi \\ \Delta \omega_z \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 7.14 \times 10^{-4} & 0 \\ 0 & 0 \\ 0 & 0.001 \end{pmatrix} \begin{pmatrix} M_x \\ M_z \end{pmatrix}$$

$$\begin{pmatrix} \Delta \phi_0 \\ \Delta \theta_0 \\ \Delta \phi_0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.3 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \Delta \phi_f \\ \Delta \theta_f \\ \Delta \phi_f \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \Delta \omega_{x0} \\ \Delta \omega_{y0} \\ \Delta \omega_{z0} \end{pmatrix} = R_0 \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 5.27 \times 10^{-4} \\ -1.49 \times 10^{-4} \\ 1.63 \times 10^{-4} \end{pmatrix}, \quad \begin{pmatrix} \Delta \omega_{xf} \\ \Delta \omega_{yf} \\ \Delta \omega_{zf} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

① $\begin{pmatrix} \Delta \theta(t_f) \\ \Delta \omega_y(t_f) \end{pmatrix} = \begin{pmatrix} 0 \\ \Delta \omega_y(t_f) \end{pmatrix} = e^{At_f} \begin{pmatrix} 0.3 \\ \Delta \omega_y(0^+) \end{pmatrix} = e^{At_f} \begin{pmatrix} \Delta \theta(0) \\ \Delta \omega_y(0^+) \end{pmatrix}$ $\Delta F_y(0) = \Delta \omega_y(0^+) - \Delta \omega_y(0) = -2.34 \times 10^{-3} \text{ (rad/s)}$
 ② $\Delta F_y(t_f) = \Delta \omega_y(t_f) - \Delta \omega_y(t_f^-) = 2.50 \times 10^{-3} \text{ (rad/s)}$
 그림은 5/17 2-d

③ $\begin{pmatrix} \Delta \phi(t_f) \\ \Delta \omega_x(t_f) \\ \Delta \phi(t_f) \\ \Delta \omega_z(t_f) \end{pmatrix} = \begin{pmatrix} 0 \\ \Delta \omega_x(t_f) \\ 0 \\ \Delta \omega_z(t_f) \end{pmatrix} = e^{At_f} \begin{pmatrix} 0 \\ \Delta \omega_x(0^+) \\ 0.5 \\ \Delta \omega_z(0^+) \end{pmatrix} = e^{At_f} \begin{pmatrix} \Delta \phi(0) \\ \Delta \omega_x(0^+) \\ \Delta \phi(0) \\ \Delta \omega_z(0^+) \end{pmatrix}$
 $\Delta F_x(0) = \Delta \omega_x(0^+) - \Delta \omega_x(0) = -1.44 \times 10^{-4} \text{ (rad/s)}$
 $\Delta F_x(t_f) = \Delta \omega_x(t_f) - \Delta \omega_x(t_f^-) = -1.44 \times 10^{-4} \text{ (rad/s)}$
 $\Delta F_z(0) = \Delta \omega_z(0^+) - \Delta \omega_z(0) = -4.31 \times 10^{-3} \text{ (rad/s)}$
 $\Delta F_z(t_f) = \Delta \omega_z(t_f) - \Delta \omega_z(t_f^-) = 4.16 \times 10^{-3} \text{ (rad/s)}$
 그림은 5/17 2-b

④ $\begin{pmatrix} \Delta \theta(t_f) \\ \Delta \omega_y(t_f) \end{pmatrix} = e^{At_f} \begin{pmatrix} \Delta \theta(0) \\ \Delta \omega_y(0) \end{pmatrix} + \int_0^{t_f} e^{A(t_f-s)} \cdot B \cdot \bar{M}_y \cdot ds + \int_{t_f}^{t_f} e^{A(t_f-s)} \cdot B \cdot -\bar{M}_y \cdot ds$
 $\begin{pmatrix} t_f \\ \bar{M}_y \end{pmatrix} = \begin{pmatrix} 59.07 \text{ s} \\ -0.137 \text{ Nm} \end{pmatrix}$
 그림은 5/17 2-c

⑤ $\begin{pmatrix} \Delta \phi(t_f) \\ \Delta \omega_x(t_f) \\ \Delta \phi(t_f) \\ \Delta \omega_z(t_f) \end{pmatrix} = e^{At_f} \begin{pmatrix} \Delta \phi(0) \\ \Delta \omega_x(0) \\ \Delta \phi(0) \\ \Delta \omega_z(0) \end{pmatrix} + \int_0^{t_f} e^{A(t_f-s)} \cdot B_1 \cdot \bar{M}_x \cdot ds + \int_{t_f}^{t_f} e^{A(t_f-s)} \cdot B_1 \cdot -\bar{M}_x \cdot ds + \int_0^{t_f} e^{A(t_f-s)} \cdot B_2 \cdot \bar{M}_z \cdot ds + \int_{t_f}^{t_f} e^{A(t_f-s)} \cdot B_2 \cdot -\bar{M}_z \cdot ds$
 $\begin{pmatrix} t_s^x \\ \bar{M}_x \\ t_s^z \\ \bar{M}_z \end{pmatrix} = \begin{pmatrix} 79.72 \text{ s} \\ 2.87 \times 10^{-2} \text{ Nm} \\ 59.35 \text{ s} \\ -1.35 \times 10^{-1} \text{ Nm} \end{pmatrix}$
 그림은 5/17 2-d

3. compare 장단점 of your three different τ to rest to rest slowing down 문제.

3-1 비선형제어 문제1 3-2 선형제어 문제2-a,b 3-3 선형제어 문제2-c,d.

3-1 비선형제어는 시스템의 행동을 정확히 알려준다. 대신 계산이 오래 걸려서 시간이 많이 소요된다. 이는 비선형은 공식화하기 어렵기 때문이다.

3-2 선형제어에서 τ two impulse는 간단하고 쉽다. 어떤 선형이 있고 비선형이 없기에, 시스템의 행동을 완전 정확하게는 못하다. 이는 이론과 실제의 차이 때문이다.

3-3 3-2와 같다. piecewise constant.

Nonlinear control, Two impulse with linear, piecewise constant.

